SURIYAPRAKASH BALAMURUGAN

TECHNOLOGY LEAD | CLOUD & DATA ARCHITECT | DATA ENGINEERING & SOLUTIONS

suriyaprakash97@gmail.com +91 8526655858 linkedin.com/in/suriyaprakash-balamurugan

Pune, India | Open to Global Relocation | Requires Visa Sponsorship for International Roles

PROFESSIONAL SUMMARY

Technology Lead with **7+ years of experience** across **Data Engineering, Cloud & Solution Architecture, and Technical Leadership**, delivering scalable cloud-native data platforms for enterprise and regulated environments. Strong hands-on expertise in **AWS, Databricks, Snowflake, Spark, PySpark, Python, Terraform, Airflow, and CI/CD**, with expertise across **GenAI, RAG, Agentic AI, and MCP**. Proven track record of leading **cross-functional teams of 9+ members**, driving **40%+ efficiency gains**, and contributing to **$700K+ in cumulative savings** through cloud, workload, and infrastructure optimisation. Proven ability to translate complex business requirements into **scalable architecture, governed data platforms, and production-ready solutions** across Clinical, Product, Analytics, Data Science, and Engineering functions.

CORE COMPETENCIES

Data & Cloud Architecture | Cross-Functional Management | Stakeholder Management | Data Engineering & ETL/ELT | GenAI & AI Architecture | Cloud-Native Engineering | Technical Leadership & Governance | Data Quality | Data Governance | DevOps & CI/CD

TECHNICAL SKILLS

- **Data Engineering**: Data Engineering, ETL/ELT, Data Pipelines, Data Modelling, Data Warehousing, Data Quality, Data Lineage, Distributed Systems
- **Cloud & Data Platforms**: **AWS**, **Snowflake**, **Databricks,** Delta Lake, AWS Glue, S3, EMR, Lambda, Databricks SQL
- **Programming & Processing**: **Python**, **SQL**, PySpark, **Spark**, Snowpark, TypeScript, Node.js
- **Architecture**: **Cloud Architecture**, **Solution Architecture**, Data Architecture, Lakehouse Architecture, Cloud-Native Architecture, Platform Architecture, Enterprise Architecture, Distributed Systems
- **GenAI & AI Architecture**: **RAG**, Retrieval-Augmented Generation, **Agentic AI**, **MCP (Model Context Protocol)**, LLM Applications, MLflow, GitHub Copilot, AI-Assisted Development, Genie Space
- **DevOps & Infrastructure**: **Terraform**, Infrastructure as Code, **CI/CD**, Automated Testing, Static Code Analysis, Release Automation, Monitoring & Observability
- **Leadership**: **Technical Leadership**, Team Leadership, Architecture Governance, Mentoring, Stakeholder Management, Agile Delivery, Roadmap Planning
- **Governance & Optimisation**: GxP, Data Governance, Compliance, Auditability, Traceability, Cloud Cost Governance, Cost Optimisation

PROFESSIONAL EXPERIENCE

Business Technology Solutions Consultant (Technology Lead – Data Platforms) 03/2023 – Present

ZS ASSOCIATES Pune

- Lead **architecture and delivery** of enterprise cloud data platforms supporting Clinical R&D and regulated analytics.
- Manage and mentor a **9-member cross-functional team** across Data Engineering, Data Science and UI/UX.
- Define **architecture standards, code quality frameworks and release governance**, improving platform reliability by **35%**.
- Partner with Product, Clinical and Analytics stakeholders to translate business requirements into scalable technical solutions.
- Improved resource utilisation by **40%+** and delivered **$500K+ annual cloud savings** through workload optimisation and cost governance.
- Increased engineering velocity by **25%** through governed adoption of **GenAI, GitHub Copilot and LLMs**.
- Automated CI/CD and deployment workflows, improving engineering productivity by **70%**.
- Architect **AWS-Snowflake and Databricks Lakehouse** platforms for structured and unstructured data.
- Provision secure, scalable AWS infrastructure using **Terraform / Infrastructure as Code**.
- Build scalable ingestion, transformation and orchestration pipelines using **Spark, PySpark, Airflow and AWS**.
- Deliver secure, compliance-ready data products supporting **self-service analytics** for global business users.

Technology Analyst 09/2019 – 03/2023

INFOSYS LIMITED Chennai

- Designed and developed **cloud-native applications** using AWS, Python, TypeScript, Node.js and CI/CD.
- Led migration of legacy **DB2 systems to cloud-native architectures**, reducing operational overhead by **20%**.
- Engineered high-throughput, fault-tolerant systems delivering **40% performance improvement**.
- Standardised **CI/CD pipelines**, improving release stability and deployment velocity.
- Managed **Unit, SIT and UAT testing** across end-to-end delivery cycles.
- Mentored junior engineers on **cloud development, data engineering and technical problem-solving**.

SELECTED ARCHITECTURE PROJECTS

INSIGHT360

ZS Associates | Technology Lead 11/2024 – Present

- Designed **Databricks Lakehouse architecture** enabling self-service clinical analytics for global Medical Affairs teams.
- Built automated ingestion and harmonisation workflows for **structured and unstructured clinical data**.
- Established reusable data product patterns using **Delta Lake, Unity Catalog and MLflow** for lineage and reproducibility.
- Led **architecture decisions, backlog prioritisation and MVP roadmap** across Clinical, Data Science and Engineering teams.
- **Tech**: Databricks, Delta Lake, PySpark, Databricks SQL, Unity Catalog, MLflow, Python, Workflows, Genie Space.

DISTILL

ZS Associates | Technology Lead 04/2024 – Present

- Architected a **GxP-aligned pipeline orchestration and monitoring framework** for regulated clinical workloads.
- Implemented automated validation gates across **data quality, compliance and deployment readiness**, reducing manual QA by **65%**.
- Scaled pipelines to **5× higher data volumes** with **40% efficiency improvement** through Snowflake and Spark optimisation.
- Reduced infrastructure spend by **$200K+ annually** through AWS cost governance, compute tuning and resource auto-scheduling.
- **Tech**: Python, Snowflake, Airflow, AWS, Spark, PySpark.

VAW HYGIENE

ZS Associates | Senior Data Engineer 08/2023 – 03/2024

- Automated **95% of data ingestion and pipeline creation** using PySpark and Snowpark, improving engineering productivity by **76%**.
- Optimised SQL execution and Snowflake warehouse clustering, reducing compute consumption by **44%**.
- **Tech**: PySpark, Snowflake, Snowpark, Dataiku, AWS.

CERTIFICATIONS

Claude Certified Architect Professional | **Snowflake SnowPro Advanced: Architect** | **Snowflake SnowPro Core**

Databricks Certified Generative AI Engineer Associate | **AWS Certified Data Engineer** Associate | **SAFe 6** Practitioner

AWS Certified Developer Associate | **AWS Certified Solutions Architect** Associate | **Dataiku Core Designer** Certification

EDUCATION

Bachelor of Engineering - Mechatronics

Anna University 2015 – 2019

LANGUAGES

English	Tamil	German	Spanish
Native or Bilingual Proficiency	*Native or Bilingual Proficiency*	*Elementary Proficiency*	*Elementary Proficiency*